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Rigid body mechanics

Problem Set II

Exercise 1:

Determine the number of degrees of freedom of the following systems:

1. Two material points M_1 and M_2 moving freely on a plane while maintaining a constant distance between them.
2. A sphere free to move in space.
3. A sphere free to move on a plane.
4. A sphere free to move on a line.
5. A rod free to move on a plane.
6. A rod with one fixed end moving on a plane.
7. A system of two articulated rods with one fixed end.
8. A hoop free to move on a plane.
9. A hoop free to move on a line.

Correction

The number of degrees of freedom (DOF) is calculated as: DOF = Total parameters - Constraints.

1. **Two points on a plane with constant distance:**

   Each point has 2 DOF in the plane (4 total). The constraint of constant distance removes 1 DOF.

   $$\text{DOF} = 4 - 1 = 3$$

2. **Sphere free in space:**

   Position of center: 3 DOF. Rotation about center (any axis): Since a sphere is spherically symmetric, only 2 independent rotations are distinguishable (like latitude and longitude of a marked point). However, for a general rigid body sphere: 3 DOF.

   $$\text{DOF} = 3 + 3 = 6$$

But if the sphere is perfectly homogeneous with no markings, rotations are indistinguishable: DOF = 3.

3. **Sphere on a plane:**

   Position on plane: 2 DOF. Height fixed by contact: 0 additional DOF. Rotations: 3 DOF (roll, pitch, yaw). Constraint: contact with plane removes 1 DOF (vertical position).
   $$\text{DOF} = 2 + 3 = 5$$

4. **Sphere on a line:**

   Position along line: 1 DOF. Rotations: 3 DOF.
   $$\text{DOF} = 1 + 3 = 4$$

5. **Rod on a plane:**

   Position of center: 2 DOF. Orientation in plane: 1 DOF (angle).
   $$\text{DOF} = 2 + 1 = 3$$

6. **Rod with fixed end on a plane:**

   One end is fixed: 0 DOF for position. Orientation in plane: 1 DOF (angle of rotation about fixed point).
   $$\text{DOF} = 1$$

7. **Two articulated rods with one fixed end:**

   First rod: 1 DOF (angle from fixed point). Second rod: 1 DOF (angle relative to first rod at joint).
   $$\text{DOF} = 2$$

8. **Hoop on a plane:**

   Position of center: 2 DOF. Orientation (rotation about perpendicular axis): 1 DOF. Rotation about its own axis: 1 DOF.
   $$\text{DOF} = 2 + 2 = 4$$

9. **Hoop on a line:**

   Position along line: 1 DOF. Orientation (tilt): 2 DOF. Rotation about its axis: 1 DOF.
   $$\text{DOF} = 1 + 3 = 4$$

Exercise 2:

Let S be a perfect rigid body in motion with respect to a fixed reference frame T. $\vec{\Omega}$ is the instantaneous rotation vector of S with respect to T.

Show that at a given instant, the projection of the velocities of the points of the solid on the axis $\vec{\Omega}$ is constant.

Correction

Consider two arbitrary points A and B of the rigid body S. The velocity relationship for a rigid body is:

$$\vec{V}(B/T) = \vec{V}(A/T) + \vec{\Omega} \times \overrightarrow{AB}$$

Taking the projection on the axis of $\vec{\Omega}$ (scalar product with $\vec{\Omega}$):

$$\vec{V}(B/T) \cdot \vec{\Omega} = \vec{V}(A/T) \cdot \vec{\Omega} + (\vec{\Omega} \times \overrightarrow{AB}) \cdot \vec{\Omega}$$

The triple scalar product $(\vec{\Omega} \times \overrightarrow{AB}) \cdot \vec{\Omega}$ is zero because $\vec{\Omega} \times \overrightarrow{AB}$ is perpendicular to $\vec{\Omega}$:

$$(\vec{\Omega} \times \overrightarrow{AB}) \cdot \vec{\Omega} = 0$$

Therefore:

$$\vec{V}(B/T) \cdot \vec{\Omega} = \vec{V}(A/T) \cdot \vec{\Omega}$$

This shows that the projection of the velocity of any point B on the axis $\vec{\Omega}$ equals the projection of the velocity of point A on the same axis. Since A and B are arbitrary, this projection is constant for all points of the solid at a given instant.
Physical interpretation: All points of the solid have the same velocity component along the instantaneous rotation axis.

Exercise 3:

A rectangular plate $ABCD$ moves with respect to a direct orthonormal frame $R(O, X, Y, Z)$ such that A remains coincident with point O and side AB remains in the plane XOY.

1. Determine the number of degrees of freedom of the plate.
2. What is the rotation vector $\vec{\Omega}$ of the plate with respect to R?

Correction

1. **Number of degrees of freedom:**

   A rigid body in 3D space normally has 6 DOF (3 translations + 3 rotations).

   Constraints:

   - Point A is fixed at O: removes 3 DOF (translation blocked)
   - Side AB remains in plane XOY: This constrains the orientation. The plate can only rotate about point A such that AB stays in XOY. This removes 2 rotational DOF (no rotation about axes in the XOY plane that would lift AB out).

   The plate can only rotate about an axis perpendicular to the XOY plane, i.e., about the Z axis.

   $$\text{DOF} = 6 - 3 - 2 = 1$$

2. **Rotation vector:**

   Since the plate can only rotate about the Z axis passing through O (with A at O and AB in XOY), the instantaneous rotation vector is:

   $$\vec{\Omega} = \dot{\theta}\vec{k}$$

   where θ is the angle of rotation about the Z axis and $\vec{k}$ is the unit vector along Z.

   Alternatively, if we denote the angular velocity as ω:

   $$\vec{\Omega} = \omega\vec{k}$$

Exercise 4:

A hoop S of center C and radius r moves in the plane OX_1Y_1 of a reference frame $R_1(O, X_1, Y_1, Z_1)$ while remaining in contact with the axis OX_1 at point I, $\overrightarrow{OI} = x\vec{i}_1$. The frame $R_1(O, X_1, Y_1, Z_1)$ is in motion with respect to the fixed reference frame $T_1(O_1, X_1, Y_1, Z_1)$ such that $\overrightarrow{O_1O} = y\vec{j}_1$.

Let M be a point of the solid S. We set $(\vec{i}_1, \overrightarrow{CM}) = \theta$. We attach to the solid a frame $T(C, X, Y, Z)$ with basis $(\vec{i}, \vec{j}, \vec{k})$ such that the X axis passes through point M.

1. Determine the number of degrees of freedom of the solid.
2. Express the instantaneous rotation vector of the hoop with respect to T_1.
3. Can there be pivoting of the hoop with respect to R?
4. Determine the sliding velocity of the hoop with respect to the axis OX_1.
5. Calculate $\vec{V}(I/R)$ and $\vec{V}(I/S)$ then find again the sliding velocity of the hoop with respect to the axis OX_1.
6. Calculate the acceleration $\vec{\gamma}(M/T_1)$. Deduce the accelerations $\vec{\gamma}_S(I/T_1)$ and $\vec{\gamma}_S(J/T_1)$ where J is a point diametrically opposite to I on S.

Correction

1. **Degrees of freedom:**

   The hoop moves in plane OX_1Y_1 and remains in contact with axis OX_1.

   - Position of center C: x (along X_1) and fixed at $y_C = r$ (above X_1), so 1 DOF
   - Rotation in plane: angle θ, so 1 DOF

   Also, frame R_1 moves relative to T_1: position y of O relative to O_1, so 1 DOF.

   $$\text{Total DOF} = 3 \text{ (positions: } x, y, \theta)$$

2. **Rotation vector of hoop with respect to T_1:**

   Using the composition of angular velocities:

   $$\vec{\Omega}(S/T_1) = \vec{\Omega}(S/R_1) + \vec{\Omega}(R_1/T_1)$$

   Since the hoop rotates in the X_1Y_1 plane:

   $$\vec{\Omega}(S/R_1) = \dot{\theta}\vec{k}_1$$

   Frame R_1 only translates relative to T_1 (no rotation):

   $$\vec{\Omega}(R_1/T_1) = \vec{0}$$

   Therefore:

   $$\vec{\Omega}(S/T_1) = \dot{\theta}\vec{k}_1$$

3. **Pivoting with respect to R_1:**

   Pivoting means rotation about an axis normal to the plane of motion. Since the hoop remains in plane OX_1Y_1 and $\vec{\Omega}(S/R_1) = \dot{\theta}\vec{k}_1$ is perpendicular to this plane, there is pivoting about the Z_1 axis.

   Yes, there is pivoting.

4. **Sliding velocity with respect to axis OX_1:**

   The sliding velocity is the velocity of the contact point I along the axis OX_1 in frame R_1:
$$v_{\text{slide}} = \dot{x}$$

   Since $\overrightarrow{OI} = x\vec{i}_1$ and O is fixed in R_1:
$$\vec{V}(I/R_1) = \dot{x}\vec{i}_1 + \vec{\Omega}(S/R_1) \times \overrightarrow{CI}$$

   With $\overrightarrow{CI} = -r\vec{j}_1$ (from center to contact point):
$$\vec{\Omega}(S/R_1) \times \overrightarrow{CI} = \dot{\theta}\vec{k}_1 \times (-r\vec{j}_1) = r\dot{\theta}\vec{i}_1$$

   For rolling without slipping: $\vec{V}(I/R_1) = \vec{0}$, giving $\dot{x} + r\dot{\theta} = 0$, so $\dot{x} = -r\dot{\theta}$.

   The sliding velocity is:
$$v_{\text{slide}} = \dot{x}$$

5. **Velocities $\vec{V}(I/R_1)$ and $\vec{V}(I/S)$:**

   Point I as part of the solid S:
$$\vec{V}(I/S) = \vec{0}$$

   (by definition, I is at rest in S)

   Actually, I is the instantaneous contact point, so:
$$\vec{V}(I/S) = \vec{V}(C/R_1) + \vec{\Omega}(S/R_1) \times \overrightarrow{CI}$$

   With $\vec{V}(C/R_1) = \dot{x}\vec{i}_1$ and $\overrightarrow{CI} = -r\vec{j}_1$:
$$\vec{V}(I/S) = \dot{x}\vec{i}_1 + \dot{\theta}\vec{k}_1 \times (-r\vec{j}_1) = \dot{x}\vec{i}_1 + r\dot{\theta}\vec{i}_1 = (\dot{x} + r\dot{\theta})\vec{i}_1$$

   The sliding velocity is the component of $\vec{V}(I/R_1)$ along $\vec{i}_1$:
$$v_{\text{slide}} = \dot{x} + r\dot{\theta}$$

6. **Acceleration $\vec{\gamma}(M/T_1)$:**

   Using the general formula:
$$\vec{\gamma}(M/T_1) = \vec{\gamma}(C/T_1) + \frac{d\vec{\Omega}}{dt}(S/T_1) \times \overrightarrow{CM} + \vec{\Omega}(S/T_1) \times (\vec{\Omega}(S/T_1) \times \overrightarrow{CM})$$

With $\vec{\gamma}(C/T_1) = \ddot{x}\vec{i}_1 + \ddot{y}\vec{j}_1$, $\vec{\Omega}(S/T_1) = \dot{\theta}\vec{k}_1$, and $\overrightarrow{CM} = r\vec{i}$ (where $\vec{i}$ is the radial direction to M):

In the R_1 basis: $\overrightarrow{CM} = r\cos\theta\vec{i}_1 + r\sin\theta\vec{j}_1$

The centripetal acceleration term:

$$\vec{\Omega} \times (\vec{\Omega} \times \overrightarrow{CM}) = -\dot{\theta}^2 r(\cos\theta\vec{i}_1 + \sin\theta\vec{j}_1)$$

Therefore:

$$\vec{\gamma}(M/T_1) = (\ddot{x} - r\dot{\theta}^2\cos\theta)\vec{i}_1 + (\ddot{y} - r\dot{\theta}^2\sin\theta)\vec{j}_1 + r\ddot{\theta}(-\sin\theta\vec{i}_1 + \cos\theta\vec{j}_1)$$

For point I (at $\theta = -\pi/2$, i.e., bottom of hoop):

$$\vec{\gamma}_S(I/T_1) = (\ddot{x} + r\ddot{\theta})\vec{i}_1 + (\ddot{y} + r\dot{\theta}^2)\vec{j}_1$$

For point J (at $\theta = \pi/2$, top of hoop):

$$\vec{\gamma}_S(J/T_1) = (\ddot{x} - r\ddot{\theta})\vec{i}_1 + (\ddot{y} - r\dot{\theta}^2)\vec{j}_1$$

Exercise 5:

Consider a homogeneous sphere $S(G, a)$, of center G, mass m, and radius a, moving on the fixed horizontal plane (O, X_0, Y_0) of a direct orthonormal fixed frame $R_0(O_0, X_0, Y_0, Z_0)$. Let $R_S(G, X, Y, Z)$ be the direct orthonormal frame attached to the sphere. I is the contact point between the sphere S and the plane (O, X_0, Y_0). We denote by ψ, θ, and φ the three Euler angles of S/R_0 and x, y, and z the coordinates of G. We assume that S rolls without slipping on the axis (O, X_0).

1. Euler angles

a) Identify the relative Euler frames R_1 and R_2. Deduce the angular variables.

b) Parameterize the sphere S.

c) Determine the expression of the instantaneous rotation vector of S with respect to R_0.

2. Kinematics of the contact point

a) Express the non-pivoting condition of S with respect to R_0.

b) Express the non-rolling condition of S with respect to (O, X_0). Show that one of the obtained equations is semi-holonomic.

c) Calculate the velocities of the different contact points: Point of the sphere $I \in S$; point of the material plane $I \in (O, X_0, Y_0)$; and geometric point $I \in E$.

d) Give the no-slip condition of S with respect to (O, X_0, Y_0). What is the number of degrees of freedom of S in its motion with respect to R_0?

Correction

1. Euler angles

a) **Euler frames:**

The Euler angles (ψ, θ, φ) define the orientation:

- R_1: obtained from R_0 by rotation ψ about Z_0 (precession)
- R_2: obtained from R_1 by rotation θ about the line of nodes (nutation)
- R_S: obtained from R_2 by rotation φ about Z (spin)

Angular variables: ψ (precession), θ (nutation), φ (spin)

b) **Parameterization:**

The sphere requires 6 parameters:

- Position of center G: (x, y, z) in R_0
- Orientation: Euler angles (ψ, θ, φ)

Since the sphere rolls on plane (O, X_0, Y_0): $z = a$ (constant).

Parameters: $(x, y, \psi, \theta, \varphi)$ with constraint $z = a$.

c) **Instantaneous rotation vector:**

Using Euler angles:

$$\vec{\Omega}(S/R_0) = \dot{\psi}\vec{k}_0 + \dot{\theta}\vec{u} + \dot{\varphi}\vec{k}$$

where $\vec{u}$ is the line of nodes and $\vec{k}$ is the axis of R_S.

In the R_0 frame:

$$\vec{\Omega}(S/R_0) = \dot{\psi}\vec{k}_0 + \dot{\theta}\vec{u} + \dot{\varphi}\vec{k}$$

2. Kinematics of contact point

a) **Non-pivoting condition:**

Non-pivoting means no rotation about the vertical axis at the contact point. The vertical component of angular velocity must be zero:

$$\vec{\Omega}(S/R_0) \cdot \vec{k}_0 = 0$$

This gives: $\dot{\psi} + \dot{\varphi}\cos\theta = 0$

b) **Non-rolling condition on (O, X_0):**

Rolling on axis X_0 means the sphere moves along X_0 without slipping perpendicular to it. The velocity of contact point I must be along X_0:

$$\vec{V}(I/R_0) \cdot \vec{j}_0 = 0$$

With $\vec{V}(I/R_0) = \vec{V}(G/R_0) + \vec{\Omega}(S/R_0) \times \overrightarrow{GI}$:

$$\dot{y} + [\vec{\Omega}(S/R_0) \times (-a\vec{k}_0)]_y = 0$$

This yields: $\dot{y} - a\omega_x = 0$ where ω_x is the X_0 component of $\vec{\Omega}$.

One equation is **semi-holonomic** if it involves velocities but cannot be integrated to give a position constraint (e.g., $\dot{y} = a\dot{\varphi}\sin\psi$).

c) **Velocities:**

- Point $I \in S$ (material point of sphere):

$$\vec{V}(I \in S/R_0) = \vec{V}(G/R_0) + \vec{\Omega}(S/R_0) \times \overrightarrow{GI}$$
$$= \dot{x}\vec{i}_0 + \dot{y}\vec{j}_0 + \vec{\Omega} \times (-a\vec{k}_0)$$

- Point $I \in$ plane (material point of fixed plane):

$$\vec{V}(I \in \text{plane}/R_0) = \vec{0}$$

- Geometric point $I \in E$ (space point):

$$\vec{V}(I \in E/R_0) = \vec{V}(G/R_0) = \dot{x}\vec{i}_0 + \dot{y}\vec{j}_0$$

d) **No-slip condition and DOF:**

No-slip means $\vec{V}(I \in S/R_0) = \vec{V}(I \in \text{plane}/R_0) = \vec{0}$:

$$\dot{x}\vec{i}_0 + \dot{y}\vec{j}_0 + \vec{\Omega}(S/R_0) \times (-a\vec{k}_0) = \vec{0}$$

This gives 2 constraints (in X_0 and Y_0 directions). Combined with rolling on X_0 (1 additional constraint):

Total parameters: 5 (since $z = a$). Constraints: 2 (no-slip).

$$\text{DOF} = 5 - 2 = 3$$

Exercise 6:

A horizontal disk S_2, of center O_1 and radius R, is rotated about the axis O_1Z_1 of the reference frame $T_1(O_1, X_1, Y_1, Z_1)$ assumed fixed, with angular velocity ω, by a disk S_1, of center C and radius r. S_1 rolls without slipping on the contour of disk S_2, with angular velocity Ω, about the rod AC which is parallel to the plane of S_2. This rod rotates about the axis AZ_1 with angular velocity ω_0 with respect to S_2.

1. Write the expressions for the rotation velocity vectors $\vec{\Omega}(S_1/AC)$, $\vec{\Omega}(AC/S_2)$, and $\vec{\Omega}(S_2/T_1)$.
2. By exploiting the rolling without slipping condition of S_1 on S_2, deduce a relation between the angular velocities Ω, ω_0, and ω.
3. Express the velocity of point J of S_1, which is diametrically opposite to I. What does this velocity become if disk S_2 is stationary?

Correction

1. **Rotation velocity vectors:**

   - $\vec{\Omega}(S_1/AC)$: Disk S_1 rotates about rod AC with angular velocity Ω. If $\vec{u}_{AC}$ is the unit vector along AC:
   $$\vec{\Omega}(S_1/AC) = \Omega\vec{u}_{AC}$$
   - $\vec{\Omega}(AC/S_2)$: Rod AC rotates about axis AZ_1 with angular velocity ω_0 relative to S_2:
   $$\vec{\Omega}(AC/S_2) = \omega_0\vec{k}_1$$
   - $\vec{\Omega}(S_2/T_1)$: Disk S_2 rotates about O_1Z_1 with angular velocity ω:
   $$\vec{\Omega}(S_2/T_1) = \omega\vec{k}_1$$

2. **Relation between angular velocities:**

   Using composition of angular velocities: $\vec{\Omega}(S_1/T_1) = \vec{\Omega}(S_1/AC)+\vec{\Omega}(AC/S_2)+\vec{\Omega}(S_2/T_1) = \Omega\vec{u}_{AC} + \omega_0\vec{k}_1 + \omega\vec{k}_1 = \Omega\vec{u}_{AC} + (\omega_0+\omega)\vec{k}_1$

   At the contact point I between S_1 and S_2, the rolling without slipping condition requires: $\vec{V}(I \in S_1/T_1) = \vec{V}(I \in S_2/T_1)$

   For disk S_2, point I is at radius R from O_1: $\vec{V}(I \in S_2/T_1) = \vec{\Omega}(S_2/T_1)\times\overrightarrow{O_1I} = \omega\vec{k}_1 \times R\vec{u}_r$ where $\vec{u}_r$ is the radial direction from O_1 to I.

   For disk S_1, point I is at the contact (on the rim): $\vec{V}(I \in S_1/T_1) = \vec{V}(C/T_1)+\vec{\Omega}(S_1/T_1) \times \overrightarrow{CI}$

   The center C of S_1 moves in a circle of radius R around O_1: $\vec{V}(C/T_1) = (\omega_0+\omega)\vec{k}_1 \times R\vec{u}_r$

   At the contact point, $\overrightarrow{CI}$ is perpendicular to AC with magnitude r. The no-slip condition gives: $\vec{V}(I \in S_1/T_1) = \vec{V}(I \in S_2/T_1)$

   Expanding and comparing tangential components: $(\omega_0+\omega)R - \Omega r = \omega R$

Simplifying: $\omega_0 R = \Omega r$

Therefore: $\boxed{\Omega = \frac{\omega_0 R}{r}}$

Or including the motion of S_2: $\boxed{\Omega = \frac{(\omega_0 + \omega)R - \omega R}{r} = \frac{\omega_0 R}{r}}$

3. **Velocity of point J:**

   Point J is diametrically opposite to I on disk S_1. Thus $\overrightarrow{CJ} = -\overrightarrow{CI}$.

   The velocity of J is: $\vec{V}(J/T_1) = \vec{V}(C/T_1) + \vec{\Omega}(S_1/T_1) \times \overrightarrow{CJ} = (\omega_0 + \omega)\vec{k}_1 \times R\vec{u}_r + [\Omega\vec{u}_{AC} + (\omega_0 + \omega)\vec{k}_1] \times (-\overrightarrow{CI})$

   Since $\vec{\Omega}(S_1/T_1) \times \overrightarrow{CI}$ gives the velocity contribution at I, and we have: $\vec{\Omega}(S_1/T_1) \times \overrightarrow{CJ} = -\vec{\Omega}(S_1/T_1) \times \overrightarrow{CI}$

   From the contact point analysis: $\vec{V}(I/T_1) = \vec{V}(C/T_1) + \vec{\Omega}(S_1/T_1) \times \overrightarrow{CI} = \omega R\vec{u}_\theta$

   Therefore: $\vec{V}(J/T_1) = \vec{V}(C/T_1) - [\vec{V}(I/T_1) - \vec{V}(C/T_1)] = 2\vec{V}(C/T_1) - \vec{V}(I/T_1)$ $= 2(\omega_0 + \omega)R\vec{u}_\theta - \omega R\vec{u}_\theta = [(2\omega_0 + 2\omega - \omega)R]\vec{u}_\theta$ $\boxed{\vec{V}(J/T_1) = (2\omega_0 + \omega)R\vec{u}_\theta}$

   If disk S_2 is stationary ($\omega = 0$): $\boxed{\vec{V}(J/T_1) = 2\omega_0 R\vec{u}_\theta}$

   The point J moves with twice the velocity of the center C, which is characteristic of rolling motion.